

SOP-41: Margin Table and Economics Refresh

Version 1.0 · 2026-07-28 · Owner: PPC Lead · First build against the SOP Build Kit v1.0 12-element standard · Thresholds live in the Master PPC Decision Criteria System v4.0 only; this SOP carries zero local thresholds · Due date honored: 2026-08-21, before Wave 1 opens

1. Verdict and Scope

VERDICT: every ceiling in the account is derived from the margin table, and today the margin table is one blended row. The T0 economics spine currently reads a single blended B4 figure (contribution \$28.37 on AOV \$81.05, break-even ACOS 35.0%, version Jun-2026), which means every Max Profitable CPC, every Target and Max Acceptable ACOS, every push loss ceiling, and every sibling composite input in the system is computed from an average that is wrong for every individual variation. This SOP owns the process that ends that state and keeps it ended: the monthly and event-driven refresh of the D1 per-variation margin table, its validation, its version stamping, the G9 freshness state, the repoint of the T0 economics spine from blended to variation-weighted values, and the same-day routing of ceiling recomputation to the ladder queue whenever margin moves. Completing this component clears external dependency D1 in the Cross-Reference Ledger Section 2, which unblocks component #11 at live depth and SOP-23 push sizing at real margin figures.

The refresh is a data-stewardship SOP, not a keyword-verdict SOP: it issues no keyword verdicts of its own. It produces the states (G9 fresh, PROVISIONAL, stale, quarantined) that unblock or suppress the verdict classes v4.0 issues, and it hands exactly one worklist downstream: the ladder-queue sweep that "8.1 Product-Level Scenarios (S-A)" rule S-A9 orders when ceilings recompute. Twelve procedures, eight failure modes, and four worked examples follow. The document runs the full 12-element Build Kit standard; the arithmetic in Section 6 is asserted programmatically.

1.1 Authority note for this build

The threshold authority attached to this build session carries the filename Master_PPC_Decision_Criteria_System_v4_0.pdf and internal version labels reading 3.0 on the title page and every footer. The program owner ruled on 2026-07-28 that the attached file is the governing v4.0. This SOP therefore cites it as v4.0, quoting the section headings exactly as they appear in the attached file, and the internal-label defect is logged as amendment item A5 in the closing section. Where an instrument (the KDR v3.0 template, the Command Workbook REFERENCE sheet, the Data Architecture) carries a rule ID or count that the attached authority assigns differently, the discrepancy is logged (A6) and this SOP quotes only material present verbatim in the attached authority or the KDR template, per anti-drift rule 2.

1.2 Boundaries with adjacent components

Adjacent component	The seam, stated so it is owned
Criteria System v4.0 (#8)	Decides every band, class boundary, gate rule, and ceiling formula this SOP applies: the margin-dollar classes, the return-rate bands, the 50% and 75% BE lines, the G9 price-stability window, the O3 THIN bar, the S-A9 recompute order. This SOP executes the refresh; it never sets a band.
#11 Economics worked-example set	Owns the training arithmetic cases (E1-E8). This SOP never restates a worked case; Section 6 examples demonstrate the refresh process itself, and Element 11 sends the incoming operator to #11 for the arithmetic standard.
#39 Data feed and manifest operations	Owns feed definitions and the T7 manifest field-group scheme. This SOP updates the T7 D1 row as an output; it does not define manifest structure.
SOP-23 Ranking-push operations	Consumes margin figures in M2 sizing. Push decisions and sizing execution live there; this SOP only guarantees the figure and its freshness state.
SOP-12 Exact campaign operations	Executes the campaign-level bid steps that the S-A9 ladder-queue sweep produces. This SOP hands the queue; it never touches a bid.
#33 Sibling arbitration	Consumes per-sibling margin per order with source state carried (SOP-33 Section 3). This SOP supplies the figure and its G9 state; the composite and the declaration live there.
#31 Budget and envelope	Envelope raises and cuts that a margin change may argue for are envelope decisions, made in the #31 instruments (per Ledger conflict C3, cite the instrument, never a #31 procedure number).
D2 keyword-variation map	Tab 2 of the Economics & Context Pack, owned by Logistics plus Catalog per the Pack READ ME. This SOP reads the ACTIVE-variation roster from it and nothing more; RE-POINT VARIATION execution belongs to SOP-12 P10.

#3 Authority Matrix	Owns decision rights. No v2.0 matrix row names the #41 refresh decisions; the interim assignment in Elements 9 and 11 is documented from the Context Pack ownership lines and logged as amendment item A8.
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2. Trigger Conditions

Verdict first: five triggers arm this SOP, and four of the five carry a same-day or same-session lag, because a stale ceiling misprices every click bought under it.

Trigger	Detecting artifact	Maximum lag from signal to procedure start
Monthly refresh due	D1 tab Version column vs the calendar month; T7 D1 row Refresh field reads Monthly. The monthly computation cadence is v4.0's: " <i>F1 Product Economics</i> ", contribution row: "Computed monthly from SellerBoard + finance; FIFO landed-cost basis."	P1 opens on Finance delivery; the completed refresh lands before the next Monday 09:00 packet-ready read of the weekly cycle (the calendar slot belongs to #4; this SOP lands its output ahead of it).
Margin-moving event	T5 Product Log and the Context Pack EVENT INTAKE sheet, closed event types "Price move" and "Fee/dimension change"; a COGS change notice from Finance routes through the same intake.	Same day. " <i>8.1 Product-Level Scenarios (S-A)</i> " S-A9: "All ceilings recompute immediately from the new margin table ... Ceilings are live values, not annual settings." P9 opens the day the event lands.
Return-rate band change	The SellerBoard true-up read against the v4.0 F1 return-rate bands (LOW <4% · NORMAL 4-8% · HIGH 8-12% · SEVERE >12%).	Same session as the true-up read; a band change re-opens P2 for the affected variation. SEVERE additionally freezes scale on the variation per F1 and routes the cause to the product team.
New ACTIVE variation	The D2 map roster and the listing registry: a variation appears in D2 or in campaign targeting with no D1 row.	Before any ceiling-referencing verdict on the variation: P10 runs ahead of the first weekly pass that reads the variation, and until the row exists the variation's actions carry the G9 PROVISIONAL tag per the SOP-12 B8 precondition.
Staleness clock expiry	T7 D1 row Status column against the 45-day version-age clock (amendment item A1: the clock value comes from Data Architecture v1.0 Section 3 and is not yet in v4.0).	Same day: P11 flips the state, opens the Finance task, and tags downstream reads. No verdict waits on the flip; the flip is what the verdicts read.

3. Preconditions and Gates

Verdict first: the refresh runs only on inputs it can validate, and a refresh that cannot validate leaves the prior version in force rather than publishing a doubtful number. Each gate below names its manifest ID from the T7 scheme (#39) and the observable that proves it passed.

Gate (manifest ID)	Requirement and passing observable
D1 delivery present (T7 row "D1 Per-variation margin")	The Finance delivery via the P1.6 CM waterfall covers every ACTIVE-status variation row in the D1 tab. Observable: zero ACTIVE rows with a blank Contribution \$/unit cell after P4 writes. Partial delivery is legal but leaves G9 PROVISIONAL per the Pack READ ME clearance rule.
Input validity (G1 pattern, "5. Layer 0: Gates (G1-G12). Nothing Proceeds Until These Pass")	Per row, the contribution identity holds: Contribution \$/unit = AOV minus COGS minus FBA fee minus referral minus freight-in (freight treated as 0 if blank, per the D1 tab header). Observable: computed value matches the delivered value within \$0.01. Two values for one metric quarantines the row per the G1 rule; the prior version's row stays in force.
Valuation consistency (Pack READ ME F1 check)	Freight-in treatment matches across valuation contexts; D1 uses landed unit economics. Observable: the delivery's cover note states the freight basis and it matches the D1 convention; a basis change is a P2 finding, never a silent absorb.
Return-rate basis stated	The delivery states whether contribution values are net of returns. The Data Architecture Section 3 file map specifies the D1 margin table as post-returns, and the live blended row derives from net revenue, so the standing convention is net-of-returns inputs; the D1 tab carries no return-rate column, which is logged as amendment item A4. Observable: the cover note's basis line reads net-of-returns.
B8 economics chain live (T7 row "B8 Row economics")	The T0 named cells (AOV, MarginPerOrder, BE_ACOS, TargetACOS, MaxAccACOS, ExpectedCPC, WeeklyEnvelope at C18:C24) resolve without error before and after the repoint. Observable: zero #REF or #VALUE reads on the spine after P6.

Completeness honesty (G8, F15)	The T7 manifest is updated before downstream analysis runs, per "F15 Verification & Data Quality": completeness classes FULL at or above 95%, WORKABLE 85-95%, DEGRADED below 85% of gate-relevant fields, with DEGRADED producing flags only. Observable: the T7 D1 row Status and the completeness score reflect this refresh before the Monday cascade opens.
Price stability awareness (G9, same gates section)	Any price change on a variation within the trailing 7 days marks that variation's CVR and ACOS reads PROVISIONAL per the G9 rule. The refresh recomputes ceilings the same day regardless: the ceiling depends on margin, not on a settled CVR read. Observable: the refresh log entry lists which variations carry the G9 price flag at write time.

4. Step-Level Procedures

Verdict first: twelve procedures, P1-P12. P1-P8 are the monthly spine; P9-P11 are the off-cycle paths; P12 closes every run. Time estimates assume one product with up to 10 ACTIVE variation rows and scale roughly linearly; they are operating estimates, not thresholds.

P1. Monthly intake and delivery check (15 min)

1. Open the Economics & Context Pack for the product, D1 MARGIN TABLE tab. Confirm the current Version stamp and Effective Date (live example: Jun-2026, 2026-06-30 on the B4 blended row).
2. Receive the Finance delivery: the per-variation P1.6 CM waterfall output, one line per ACTIVE variation, carrying AOV \$, COGS \$/unit, FBA fee \$/unit, referral \$/unit, freight-in \$/unit, and stated contribution \$/unit, on the FIFO landed-cost basis named in v4.0 "F1 Product Economics".
3. Pull the SellerBoard monthly true-up read for the product (the financial source of truth per Data Architecture Section 12) and hold it beside the delivery for the P2 reconciliation.
4. Roster check against the D2 tab: every variation with Status ACTIVE in D1, plus every variation named as a preferred or backup SKU in D2, must appear in the delivery. A missing variation is a partial delivery: proceed on the rows received, list the missing rows in the P12 log, and leave G9 PROVISIONAL.
5. Confirm the delivery cover note states the freight-in basis and the returns basis (Section 3 gates). A note missing either line goes back to Finance the same day with the two-line request; the refresh proceeds on nothing until the basis is stated.

P2. Row validation (15 min)

1. Per row, recompute the contribution identity: AOV minus COGS minus FBA minus referral minus freight-in. Compare to the delivered contribution at \$0.01 tolerance.
2. An identity failure quarantines the row per the G1 rule in "5. Layer 0: Gates (G1-G12). Nothing Proceeds Until These Pass": the row does not write, the prior version's row (or the blended row, on a first fill) stays in force for that variation, a repair task opens to Finance the same day, and the record carries the verdict string **FLAG: DATA REPAIR** from "13.1 The Closed Verdict Vocabulary".
3. Referral sanity: referral \$/unit divided by AOV should sit at the category referral rate; a ratio off the category rate is a query to Finance, not an edit. The operator never edits a delivered value; the delivery is corrected at source or the row quarantines.
4. Reconcile the delivery's product-level total against the SellerBoard true-up: variation contributions weighted by ordered units should reconcile to the SellerBoard product contribution within the noise of the month's mix. A material gap is a P2 finding naming the largest contributing row.
5. Return-rate read: pull per-variation return rates from the true-up and band them per v4.0 F1 (LOW <4%, NORMAL 4-8%, HIGH 8-12%, SEVERE >12%). A band change is recorded in the row's Notes; SEVERE fires the F1 consequence (scale freeze on the variation and a listing-quality diagnosis routed to the product team) and is escalated in P12.

P3. Class banding and band-change detection (5 min)

1. Assign each validated row its margin-dollar class per v4.0 F1: THIN below \$6, STANDARD \$6-15, RICH above \$15 per unit.
2. Compare against the prior version's class. Every class change is a broadcast item for P8, because the class carries decision consequences: F1 bars THIN variations from Ranking objectives on terms whose Expected CPC exceeds margin \$ divided by 3 (the O3 realism input in "6. Layer 1: Objective Assignment (O1-O8)").
3. Record clicks-to-loss per row (margin \$ divided by reference CPC) as the row's tolerance line, per the F1 field "clicks-to-loss (margin \$ ÷ CPC)". This is display arithmetic for the operator, not a new threshold.

P4. Version stamping and D1 write (10 min)

1. Write validated rows into the D1 tab: one row per variation, blue input cells only, formulas untouched. Contribution \$ and % compute in-sheet; confirm they match the P2 values.
2. Stamp every written row with the same Version (month label) and Effective Date. Mixed stamps inside one refresh are illegal: a refresh is atomic per product.

3. The blended row stays in the tab, marked as blended, for the reconciliation trail; after the first full per-variation fill it is never again the spine source.
4. Never fork the table. One editable D1 exists per product, in the Economics & Context Pack, per Data Architecture design rule 2 (one source of truth per field). A margin figure appearing in any second editable location is failure mode F3.

P5. G9 state evaluation (5 min)

The state resolves by this tree, in order; first match wins:

1. **QUARANTINE branch:** any ACTIVE row failed P2 identity this run: those variations stay on the prior version, the product-level G9 state stays PROVISIONAL, and the repair task carries the release condition.
2. **CLEAR branch:** every ACTIVE variation row carries Contribution \$/unit and a Version stamp, no quarantines open, version age inside the staleness clock: G9 clears per the Pack READ ME rule ("G9 flips from PROVISIONAL when every ACTIVE variation row carries Contribution \$/unit and a Version stamp"), and the spine repoints in P6.
3. **PROVISIONAL branch:** at least one ACTIVE row is blank (partial delivery, new variation pending): G9 stays PROVISIONAL; ceiling-referencing actions carry the tag per the SOP-12 B8 precondition; the missing rows are named in T7.
4. **STALE branch:** no delivery and version age past the 45-day clock (A1): P11 governs; the T7 D1 row flips to STALE and the escalation opens.

P6. Spine repoint (10 min)

1. On G9 CLEAR, repoint the T0 economics spine named cells (C18:C24) from the blended row to variation-weighted values, per the Pack READ ME: "the Command Workbook economics spine repoints from the blended row to variation-weighted values."
2. Weighting basis: trailing 30-day ordered units per variation from the SellerBoard true-up. The corpus fixes the repoint but not the weighting basis; the basis is proposed here and logged as amendment item A2 for v4.1 adoption. The basis used is written into the spine's source note every refresh.
3. Recompute the M1 chain from the new MarginPerOrder and AOV per "10. Layer 5: Sizing Mathematics (M1-M10). Every Action Carries Numbers": break-even $ACOS = \text{margin} \div \text{AOV}$; Target = 50% of BE; Max Acceptable = 75% of BE. These are formula cells; the check is that they moved.
4. Per-variation ceilings flow to the T2 K6 economics block through the D1 link (Data Architecture Section 12, Economics & Context Pack feed row): confirm one oversight-class keyword row shows the new per-variation Margin \$/unit for its targeted variation.
5. Update the T0 zone C "Margin table version" line and its G9 state display in the same session; a spine repoint with a stale version line is failure mode F1.

P7. Ceiling recompute and ladder-queue sweep (20 min)

1. Execute the S-A9 order from "8.1 Product-Level Scenarios (S-A)": "All ceilings recompute immediately from the new margin table; every keyword whose CPC is above its new ceiling enters the ladder queue this cycle."
2. Filter T2 to rows where blended CPC exceeds the row's new Max Profitable CPC (margin \$ of the targeted variation multiplied by the row's CVR, the F1 formula). Read each row's CPC ratio band per F1: at or below 0.8× ceiling SAFE, 0.8-1.0× AT-CEILING, 1.0-1.5× PUSH-ONLY, above 1.5× UNJUSTIFIABLE outside a sized push.
3. Rows above ceiling with a live M2-sized push record stay inside their push governance (the push, not this SOP, owns them). Rows above ceiling with no sizing record enter the ladder queue with the verdict class **BID DOWN TO PROFITABLE CPC; CAP SYNTAX** quoted from "13.1 The Closed Verdict Vocabulary"; issuance is v4.0's and execution is SOP-12's; this SOP produces the queue list only.
4. Hand the queue list to the weekly pass with the refresh version stamp on it, so every resulting bid step's reason string can name the margin version that triggered it.

P8. Class-flip and eligibility broadcast (10 min)

1. For every P3 class change, notify the objective owner of the consequence in one line each: a flip into THIN arms the F1/O3 bar (no Ranking objective where Expected CPC exceeds $\text{margin} \$ \div 3$); a flip out of THIN releases it; a return-rate move into SEVERE freezes scale on the variation per F1.
2. Terms on a newly THIN variation that currently carry a Ranking objective are flagged to the weekly pass for objective review under O3; the review and any transition are v4.0 and SOP-12 territory.
3. Sibling composites: notify #33 that per-sibling margin inputs changed, with the G9 state carried, per SOP-33 Section 3; re-arbitration on a margin driver is SOP-33 P6's call, not this SOP's.

P9. Event-driven off-cycle refresh (25 min)

1. Trigger: a T5 or EVENT INTAKE row of type Price move, Fee/dimension change, or a Finance COGS notice touching an ACTIVE variation.
2. Run P2-P7 for the affected variations only, same day, per the S-A9 immediacy rule. Unaffected rows keep their stamp; the affected rows take an event-dated stamp with the event reference in Notes.
3. Price moves additionally arm G9 price stability for the trailing 7 days on the variation: CVR and ACOS reads PROVISIONAL per the G9 rule, while the ceiling recompute proceeds the same day. The two clocks are different on purpose: the ceiling is arithmetic on margin, the rate read needs settled behavior.
4. If the event landed late (found in a weekly read rather than routed within the Pack's 24-hour intake window), count and log the days late in the P12 entry and route the intake miss as a finding to the intake owner.

P10. New-variation onboarding (10 min)

1. Trigger: a variation reaches ACTIVE in the listing registry or appears in D2 as a preferred or backup SKU with no D1 row.
2. Open the D1 row with identity fields (SKU, size, color, Status) the same day; request the Finance line for it in the next delivery or immediately if campaigns will target it sooner.
3. Until the row fills, every ceiling-referencing action on the variation carries the G9 PROVISIONAL tag (the SOP-12 B8 precondition), and the T7 D1 row names the pending variation. The variation is never given the blended figure as if it were its own.

P11. Staleness watch and escalation (5 min weekly)

1. Weekly, inside the Monday manifest update: compute version age = today minus Effective Date. At age past 45 days (A1), flip the T7 D1 Status to STALE, tag the product's ceiling-referencing verdict classes PROVISIONAL, and open the Finance task same day with the age counted.
2. STALE never blocks the weekly pass; it changes what the pass may claim. Per G8 and F15, if the D1 absence pushes gate-relevant completeness below 85%, verdicts on economics-dependent rows downgrade to flags and the analysis states what it could not see.
3. The state releases only through a completed P1-P6 run; there is no manual un-stale.

P12. Record close (10 min)

1. T7 manifest: update the D1 row (Status, note naming any missing variations or quarantines) and let the completeness score recompute before the cascade reads it.
2. T6 decision log: one entry per refresh carrying prior and new spine values (MarginPerOrder, BE_ACOS), the version stamp, the count of ladder-queue rows handed to P7's consumer, the class flips broadcast, and the quarantine list. The refresh entry is a record, not a prediction; the ladder-queue entries it spawns carry their own predictions where SOP-12 executes them.
3. KDR header: the cycle file generated after the refresh carries the new Margin Table Ver. value in its Gates block (the KDR template's Margin Table Ver. column), so every frozen record names the margin version it was judged under.
4. Escalations: SEVERE return band, repeated quarantines from the same source, and any second consecutive missed monthly delivery escalate to the PPC Lead with the T6 history attached.

5. Decision Points: Verdict and State Mapping

Verdict first: this SOP issues states, not keyword verdicts; the table maps each refresh outcome to the gate state it sets and the verdict classes it unblocks or suppresses, with every verdict string quoted verbatim and sourced. Zero thresholds are set in this section or anywhere in this SOP.

Refresh outcome	State set	Verdict classes affected (strings verbatim, with source)
Full validated fill, all ACTIVE rows stamped	G9 CLEAR (fresh); spine variation-weighted	Ceiling-referencing classes shed the PROVISIONAL tag; RANK PUSH: FUND SIZED PUSH (KDR v3.0 closed vocabulary list) prices from real per-variation margin in M2 sizing.
Partial delivery	G9 PROVISIONAL, missing rows named in T7	Ceiling-referencing actions carry the tag (SOP-12 B8 precondition); nothing halts, everything is honest about its basis.
Row identity failure	Row quarantined; prior version governs that variation	FLAG: DATA REPAIR , quoted from v4.0 "13.1 The Closed Verdict Vocabulary": repair task with owner and date, no recommendation on the row's economics.
Impossible or conflicting source read (two values for one metric across delivery and true-up)	G1-pattern quarantine at intake	QUARANTINE: DATA REFRESH , quoted from the KDR v3.0 Bands & Verdicts Reference closed vocabulary: data-refresh task same day, no economics publish.
Ceilings recomputed with rows	Ladder queue populated (S-A9)	BID DOWN TO PROFITABLE CPC; CAP SYNTAX , quoted from v4.0 "13.1

above new ceiling and no sizing record		<i>The Closed Verdict Vocabulary</i> ”; issued by v4.0, executed by SOP-12, listed by this SOP.
Inventory-driven target change discovered during roster check	Out of scope; routed	RE-POINT VARIATION (v4.0 “13.1 <i>The Closed Verdict Vocabulary</i> ”) belongs to SOP-12 P10 on D2's instruction; this SOP only guarantees the destination variation has a margin row.
Version age past the clock	STALE; PROVISIONAL tags on ceiling classes	Per G8/F15, DEGRADED gate-relevant completeness downgrades affected verdicts to flags; the analysis states what it could not see.

6. Worked Examples

Verdict first: four examples, one clean, three edged, all internally consistent and asserted in code at \$0.01 and 0.1 percentage-point tolerances. Tier 3 handling: the before-state uses the live blended B4 spine from the attached Economics & Context Pack (AOV \$81.05, contribution \$28.37, BE ACOS 35.0%, version Jun-2026); every per-variation figure is illustrative, because no live per-variation data exists yet, which is precisely the gap this SOP closes. Order-share weights (38% / 44% / 18%) are illustrative.

WE-1. Clean monthly refresh: the first unblended fill

Setup. Finance delivers three ACTIVE variation rows (illustrative). KING: AOV \$96.00, COGS \$34.20, FBA \$14.10, referral \$14.40, freight \$2.10. QUEEN: AOV \$78.00, COGS \$30.10, FBA \$12.60, referral \$11.70, freight \$1.90. TWIN: AOV \$52.00, COGS \$25.60, FBA \$10.90, referral \$7.80, freight \$1.90. Cover note states landed basis, net-of-returns. Trailing 30-day ordered-unit shares: KING 38%, QUEEN 44%, TWIN 18%.

P2 identities. KING: $96.00 - 34.20 - 14.10 - 14.40 - 2.10 = \31.20 (32.5% of AOV). QUEEN: $78.00 - 30.10 - 12.60 - 11.70 - 1.90 = \21.70 (27.8%). TWIN: $52.00 - 25.60 - 10.90 - 7.80 - 1.90 = \5.80 (11.2%). All three match the delivered values; referral checks at 15.0% of AOV on all three. Return rates 3.8%, 4.6%, 5.1%: bands LOW, NORMAL, NORMAL per F1.

P3 classes. KING RICH (\$31.20 > \$15), QUEEN RICH (\$21.70), TWIN THIN (\$5.80 < \$6). TWIN's flip into THIN arms the O3 bar at Expected CPC above $5.80 \div 3 = \$1.93$.

P5-P6. All ACTIVE rows stamped Jul-2026: G9 CLEARS. Spine repoints. Weighted contribution = $31.20 \times 0.38 + 21.70 \times 0.44 + 5.80 \times 0.18 = 11.856 + 9.548 + 1.044 = \22.45 . Weighted AOV = $96 \times 0.38 + 78 \times 0.44 + 52 \times 0.18 = \80.16 . New BE ACOS = $22.45 \div 80.16 = 28.0\%$ against the blended 35.0%: the blended row had every product-level ceiling 25% too loose ($35.0 \div 28.0 = 1.25$). Target = 14.0%; Max Acceptable = 21.0% per M1.

P7 per-variation ceilings at CVR 5.2% (F1: max profitable CPC = margin \$ × CVR). Blended: $28.37 \times 0.052 = \$1.48$. KING: $31.20 \times 0.052 = \$1.62$. QUEEN: $21.70 \times 0.052 = \$1.13$. TWIN: $5.80 \times 0.052 = \$0.30$. The blended figure was too tight for KING and 4.9 times too loose for TWIN: one number, wrong in both directions, which is the whole case for this SOP. A TWIN-targeted term at CPC \$0.72 with no sizing record reads $0.72 \div 0.30 = 2.4 \times$ ceiling, UNJUSTIFIABLE band, and enters the ladder queue.

WE-2. Edge: fee event, then a delivery that fails the identity check

Setup. A T5 Fee/dimension change lands on KING: FBA fee moves \$14.10 to \$16.85. P9 opens same day. Finance's corrected row states contribution **\$29.90**.

P2 catches it. Identity: $96.00 - 34.20 - 16.85 - 14.40 - 2.10 = \28.45 . Stated \$29.90 differs by \$1.45: two values for one metric. The row quarantines per the G1 pattern; the Jul-2026 KING row stays in force; **FLAG: DATA REPAIR** opens to Finance same day. No spine change publishes on a doubtful number.

Repair and completion. Finance confirms \$28.45 (the stated figure had the old fee). KING rewrites at \$28.45 (29.6%), still RICH, event-dated stamp. KING ceiling at CVR 5.2% = $28.45 \times 0.052 = \$1.48$. Spine reweights: $28.45 \times 0.38 + 21.70 \times 0.44 + 5.80 \times 0.18 = \21.40 ; BE = $21.40 \div 80.16 = 26.7\%$.

P7 consequence. Keyword "bamboo sheets king" runs CPC \$1.85 with no sizing record: $1.85 \div 1.48 = 1.25 \times$ ceiling, the F1 PUSH-ONLY band with no push attached. S-A9 puts it in the ladder queue under **BID DOWN TO PROFITABLE CPC; CAP SYNTAX**; SOP-12 executes the step. The lesson: the fee moved on a Tuesday, the ceiling moved the same Tuesday, and the queue entry names the margin version that moved it.

WE-3. Edge: the staleness clock does its job

Setup. Version stamp Jun-2026, Effective Date 2026-06-30. Finance misses the July delivery. The clock (A1, 45 days) runs out on 2026-06-30 + 45 = **2026-08-14**.

P11 on 2026-08-14. T7 D1 Status flips STALE; every ceiling-referencing verdict class on the product carries PROVISIONAL; the Finance task opens with age 45 counted; the weekly pass proceeds, stating on economics-dependent rows what it could not see. If D1's absence drags gate-relevant completeness under 85%, those verdicts downgrade to flags per G8/F15. On the second consecutive missed monthly, P12 escalates to the PPC Lead with the T6 history. Nothing in the account is priced against a number the system knows is 45 days old without the record saying so: that is the entire function of the state.

WE-4. Edge: a reprice flips a class and two clocks run at different speeds

Setup. QUEEN takes a permanent reprice \$78.00 to \$69.00 (T5 Price move). P9 opens same day. Referral recomputes at 15.0%: \$10.35.

P2. Contribution = 69.00 – 30.10 – 12.60 – 10.35 – 1.90 = **\$14.05** (20.4%). Identity holds.

P3/P8. Class flips RICH to STANDARD (\$14.05 inside \$6-15): broadcast to the objective owner; no O3 bar (not THIN), but the class line every sizing read carries just changed. New QUEEN ceiling at CVR 5.2% = $14.05 \times 0.052 =$ **\$0.73** against the prior \$1.13: every QUEEN-targeted term between \$0.73 and \$1.13 is now above ceiling and sweeps into the queue absent a sizing record.

The two clocks. Ceilings recompute the same day per S-A9. The CVR and ACOS reads on QUEEN terms carry the G9 price-stability PROVISIONAL tag for the trailing 7 days per the G9 gate rule, so no conversion verdict is issued on the unsettled week. Same event, two different clocks, both by rule. Spine reweights: $28.45 \times 0.38 + 14.05 \times 0.44 + 5.80 \times 0.18 =$ **\$18.04**; weighted AOV = $96 \times 0.38 + 69 \times 0.44 + 52 \times 0.18 =$ **\$76.20**; BE = $18.04 \div 76.20 =$ **23.7%**.

7. Failure Modes and Escalation

Verdict first: eight failure modes, F1-F8. The account-history motivation, where one exists, is the June 30 review's finding that ceilings ran on blended economics account-wide; F1-F4 are that finding decomposed.

Failure	Detection signal	Response
F1. Refresh lands, spine not repointed	T0 zone C margin version differs from the D1 tab Version stamp	P6 immediately; every verdict issued in the gap is re-read against the new ceilings; the gap's length in days is logged. Motivating incident: the standing blended spine this SOP exists to retire.
F2. Ceilings recomputed, ladder sweep skipped	Ladder-queue handoff count 0 in the T6 refresh entry while T2 shows rows above new ceiling without sizing records	P7 re-runs; the sweep list regenerates with the same version stamp; S-A9 is quoted in the entry so the skip cannot recur as interpretation.
F3. Parallel margin copy	A margin figure editable anywhere outside the D1 tab (a pasted table in a working file, a hardcoded spine cell)	Delete the copy, restore the reference read, per Data Architecture design rule 2; the copy's origin routes to change control. A field in two editable places is a build defect by rule.
F4. Stale margin, untagged verdicts	T7 D1 Status STALE or PROVISIONAL while ceiling-referencing records carry no flag	Re-tag the affected records this cycle; the J7-class honesty rule governs (flags state fallbacks in the words the engine computed); the tagging gap routes as a process finding.
F5. Freight basis mismatch	P2 identity fails systematically in one direction across rows, or the cover note basis line changes between deliveries	Quarantine the delivery, not the rows one by one; reconcile the valuation contexts with Finance per the Pack READ ME F1 consistency check before anything writes.
F6. Pre-returns margin used	Delivery contribution reconciles to gross rather than net revenue in the P2 SellerBoard reconciliation	Return to Finance for the net basis; the returns adjustment precedes every ceiling per F1 ("Adjusts margin to reality before any ceiling is computed"); log under A4 until the schema carries the column.
F7. Class flip not broadcast	A THIN-class variation carries a Ranking objective on a term with Expected CPC above margin $\$ \div 3$ (the F1/O3 bar) after a refresh that flipped the class	P8 broadcast re-runs; the term routes to objective review under O3; the miss is scored against the refresh entry that skipped it.
F8. Variation live with no	A variation appears in campaign targeting or D2	P10 same day; until the row fills, the variation's actions carry

margin row	with no D1 row (roster check or weekly manifest read)	PROVISIONAL and are never priced on the blended figure as their own.
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Escalation. The same failure mode twice in consecutive refreshes escalates to the PPC Lead with the T6 history. A SEVERE return band, a quarantine recurring from the same source line, and a second consecutive missed monthly delivery escalate the same day they are found.

8. Outputs and Artifacts

Verdict first: five artifacts close every run; a refresh without all five did not happen, whatever the spreadsheet says.

Artifact	Storage location	Retention	Downstream consumer
Versioned D1 rows (stamp + effective date)	Economics & Context Pack, D1 MARGIN TABLE tab	All versions retained; superseded rows archived in-tab by stamp	T0 spine, T2 K6 block, KDR Gates block, #33 A1 composite inputs, SOP-23 M2 sizing
G9 state record	T0 zone C version line + T7 D1 row	Current state live; history via T6	Every ceiling-referencing verdict class; SOP-12 B8 precondition
Refresh log entry (prior/new spine, flips, quarantines, queue count)	T6 decision log, append-only	Permanent	The weekly audit's scoring pass; escalation reviews
Ladder-queue handoff list (version-stamped)	Weekly pass worklist (T2 filter + list attached to the T6 entry)	Until executed or superseded by next refresh	SOP-12 P3/P11 execution; #31 instruments where syntax caps bind
Class-flip and eligibility notices	One-line entries to objective owners; mirrored in the T6 entry	Permanent via T6	O3 objective reviews; #33 re-arbitration triggers

9. Skill-Conversion Block

Block	Specification (SOP-specific)
Input bindings	D1 MARGIN TABLE tab (Economics & Context Pack); the Finance P1.6 CM waterfall delivery file; the SellerBoard monthly true-up export; T5 Product Log and the Pack EVENT INTAKE sheet (event types Price move, Fee/dimension change); D2 KW-VARIATION MAP roster; T0 named cells AOV, MarginPerOrder, BE_ACOS, TargetACOS, MaxAccACOS; T7 manifest D1 row; the T2 K6 economics block for the sweep. The Raw Pull Pack Additions v1.0 file was not received this session; any intake rows it defines bind here when it lands (amendment A7).
Deterministic logic	Contribution identity (AOV – COGS – FBA – referral – freight, \$0.01 tolerance); the M1 chain (BE = margin ÷ AOV, Target 50%, Max Acceptable 75%); F1 margin-dollar class banding (\$6 / \$15 boundaries) and return-rate banding (4 / 8 / 12%); the F1 ceiling formula (margin \$ × CVR) and CPC ratio bands; the Pack G9 clearance rule; the S-A9 sweep condition; the 45-day age computation (A1); the weighting formula (A2, trailing 30-day ordered units).
Human checkpoints	AI-executable with audit: identity and reconciliation checks, class banding, age computation, T7 and T6 drafting, sweep-list drafting, notice drafting. Human-mandatory: accepting any delivery that changes a published ceiling (the spine repoint confirm), quarantine release after repair, the first unblended fill per product, and every escalation send. Interim decision rights per amendment A8: the PPC Lead confirms (the Authority Matrix v2.0 carries no #41 row; its "Weekly verdicts (established US)" row, D = PPC Lead, is the nearest governing row for the downstream queue the refresh feeds); Finance owns input values per the Pack READ ME and is never overridden in-sheet.
Cadence	Monthly refresh on Finance delivery, landing before the Monday 09:00 packet-ready read; event-driven runs same day; staleness age check weekly inside the Monday manifest update; roster check every run.
Outputs	The five Section 8 artifacts; F1-F8 alarms with evidence lines (row, signal value, rule reference); the drafted Finance repair and delivery tasks.

10. Version Governance

Owner: PPC Lead. Component-specific review triggers, each forcing a re-validation of the named section before the next refresh runs on it: (1) any v4.0 change to F1, M1, G9, G8, O3, or S-A9 re-validates Sections 2-7; (2) any Economics & Context Pack schema change (a return-rate column landing per A4, a D1 column change) re-validates P2 and P4; (3) a change to the Finance P1.6 CM waterfall process or the accounting provider re-validates P1 and the Section 12 upstream contract; (4) a SellerBoard replacement re-validates every reconciliation step; (5) adoption of amendment items A1 or A2 into v4.0 retires the corresponding provisional citations here and increments this SOP; (6) the arrival of Raw Pull

Pack Additions v1.0 re-validates the Element 9 bindings (A7). Registered in the Supersession Map at publication; first document for component #41; the Ledger row returned by this build session is the ownership declaration of record.

11. Operator Ramp

Verdict first: the incoming operator earns this scope by demonstrating the arithmetic, then the process, then the judgment, on the week 1 / week 2 / week 4 ladder. Because the Authority Matrix v2.0 carries no #41 row (A8), the checkpoint holder throughout is the PPC Lead, per the matrix vacancy logic and the "Weekly verdicts (established US)" row that governs the queue this SOP feeds; the ramp updates the day a dedicated matrix row lands.

Week 1, observation and artifact familiarity. Reproduce all eight computations of the #11 worked-example set cold, per that document's own onboarding standard ("a new operator, human or agent, reproduces all eight computations cold before taking a scope"). Walk the D1 tab, the T0 spine named cells, the T7 D1 row, and one prior T6 refresh entry, and state from the artifacts alone what the current G9 state is and why. Observe one full refresh end to end.

Week 2, supervised execution. Run P1-P8 on a live monthly delivery with the PPC Lead reviewing each write before it publishes: the identity checks, the class banding, the G9 resolution, the spine repoint confirm (which stays human-mandatory regardless of operator seniority), and the sweep handoff. Run one simulated P9 event case and one P10 onboarding case against copies.

Week 4, unsupervised with audit. Execute a full refresh alone; the PPC Lead audits the T6/T7 trail afterward against the Section 8 artifact list: five artifacts present, sweep count reconciles to the T2 filter, every flip broadcast, every quarantine carrying a release condition.

Live-defense questions (reasoning, not recall): Why does the system set Max Acceptable below break-even rather than at it, and what behavior does the 25-point gap buy? Why do margin dollars cap aggressiveness independently of margin percent, and what does a two-click loss line look like in a row? Why was the blended row wrong for every variation at once, including the ones it flattered? Why do ceilings recompute the same day a price changes while conversion reads wait out the 7-day window? Why does a quarantined row leave the prior version in force instead of halting the product, and what would the halt cost that the tag does not? Why is the spine repoint human-mandatory when the arithmetic is deterministic?

12. Interface Contracts

Verdict first: seven contracts, each with the clause the corpus has been missing: what happens when the handoff does not arrive.

Direction	Sending owner	Receiving owner	Artifact and timing	On non-arrival
Upstream	Finance (P1.6 CM waterfall)	PPC Lead	Per-variation margin delivery with basis cover note; monthly stamp cadence per v4.0 F1	G9 stays or reverts PROVISIONAL; the 45-day clock (A1) runs to STALE; escalation on the second consecutive miss; ceilings hold at the last stamped version and every record says so.
Upstream	Data Operations	PPC Lead	SellerBoard monthly true-up read (financial source of truth per Data Architecture Section 12)	The refresh runs on the Finance delivery alone; the P2 reconciliation and return-rate banding defer, named as deferred in the T6 entry and the T7 note.
Upstream	Intake owner (BM / Data Ops per the Pack READ ME, 24-hour routing)	PPC Lead	T5 / EVENT INTAKE rows for Price move, Fee/dimension change, COGS notices	S-A9 fires late; the P9 run logs the days late; repeated intake misses route as findings to the intake owner; ceilings are corrected retroactively but verdicts issued in the gap are re-read.
Downstream	PPC Lead	Every workbook consumer (T0 spine, T2 K6, KDR)	Repointed named cells and version line before the Monday 09:00 packet-ready read	Failure mode F1: the cascade runs on the old spine; the pass proceeds gates-only on economics-dependent rows once detected, and the gap's verdicts re-read.
Downstream	PPC Lead	#11 and SOP-23	Cleared D1 (dependency D1 in Ledger Section 2), per-variation figures for worked cases and M2 sizing	#11 stays on the fictional training base and says so; SOP-23 sizing carries PROVISIONAL on its margin inputs; no push is sized on a figure the system cannot date.
Downstream	PPC Lead	#33 (Rollup A1)	Per-sibling margin per order with source and G9 state, on every refresh and every flip	Composites run on the last stated values with the state carried per SOP-33 Section 3; a declaration made on stale margin lists that fact in its record, and a margin-driver re-arbitration fires when the fresh figure lands.

Downstream	PPC Lead	Data Operations (T7)	D1 manifest row update each refresh, before the cascade	G8 reads a wrong manifest: the completeness score misstates what the packet can see; the correction is same day and the mis-stated cycle's affected verdicts re-read as flags.
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Closing: v4.1 Amendment Items Discovered in This Build

- A1.** The 45-day margin staleness clock exists in Master PPC Data Architecture v1.0 Section 3 (Economics & Context Pack file row: "version-dated, 45-day G9 staleness clock") but not in v4.0. Adopt the clock into v4.0 F1 or G9 so the threshold authority owns it.
- A2.** The spine repoint is specified as "variation-weighted values" (Pack READ ME) with no weighting basis fixed anywhere in the corpus. This SOP uses trailing 30-day ordered units and states the basis in every refresh note; v4.1 should fix the basis.
- A3.** The v4.0 gate heading G9 reads "Price stability (new)" while the instruments (T0 zone C, the Pack clearance rule, SOP-12 B8) apply G9 to margin freshness. Dual use of one gate ID; v4.1 should broaden the G9 definition or assign margin freshness its own ID.
- A4.** The D1 tab schema carries no return-rate column while the Data Architecture specifies post-returns contribution and v4.0 F1 requires the returns adjustment before any ceiling. Convention documented here (net-of-returns inputs, stated in the cover note); the schema is owed the column.
- A5.** The threshold authority file carries internal version labels reading 3.0 under the v4_0 filename; ruled by the program owner on 2026-07-28 to be the governing v4.0. v4.1 must correct the internal labels so anti-drift rule 2 is checkable by reading.
- A6.** Closed-vocabulary count conflict: v4.0 Section 13.1 declares eighteen actions; the KDR v3.0 template and SOP-12 v2.0 carry 25; the Data Architecture (rule 8) and the Command Workbook REFERENCE header state 26 over a 25-item validation list. This SOP quotes only strings present verbatim in v4.0 13.1 or the KDR list; v4.1 must publish one count.
- A7.** Raw Pull Pack Additions v1.0, routed to this session as Tier 2 item 3, was not attached. Any intake or feed rows it defines for the refresh are unverified; the Element 9 bindings re-validate when it lands.
- A8.** The Authority Matrix v2.0 carries no row naming the #41 refresh decisions. Interim rights documented in Elements 9 and 11 (PPC Lead confirms; Finance owns input values); a dedicated matrix row is owed at the next matrix re-issue.

Companions: Master PPC Decision Criteria System v4.0 (all thresholds and bands) · Economics & Context Pack v1.0 (the D1 instrument) · Master PPC Data Architecture v1.0 (spine, T7, feed cadence) · #11 Economics Worked-Example Set (training arithmetic) · SOP-12, SOP-23, #33, #31 (consumers, per Section 1.2) · Cross-Reference Ledger (component #41 row) · SOP Build Kit v1.0 (the 12-element standard this document is built to).